

A FOUNDATIONAL MONOGRAPH

A Theory of Finite Representation

Part I

Extent, Registration and
the Licence to Represent

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Draft note

This is the opening draft of a larger work. Its purpose is to establish a ground from which a Theory of Finite Representation can be developed without prematurely restricting the theory to a single instrument, branch of physics, mathematical controversy or historical episode. The arguments are therefore presented as an unfolding construction. Technical notation is introduced only where it performs indispensable work. The theory remains open to revision as its bridge rules, dynamics and experimental applications are developed.

The term *finite* is not used here as a casual synonym for small, bounded or computationally convenient. It indicates that an actual representation has an extent, occupies a carrier, is distinguished across an interactional window and arrives with some history. Nor does this work begin by asserting that every aspect of the Universe is known to be finite. Its narrower and more defensible starting point is that every scientific statement available to us has passed through finite acts of registration and representation.

Abstract

Science is often described as though a world of already formed objects were presented directly to an observer and then copied into mathematics. In practice, physical interaction, instrumental change, registration, numerical transformation, classification, modelling and language form an extended trajectory. This monograph begins a Theory of Finite Representation by making that trajectory explicit. It distinguishes three co-related basins: physical interaction and registration; endogenous mathematical and linguistic construction; and the bridge rules through which symbolic structures are licensed to make physical claims. The proposed Axiom of Finite Representation does not prohibit classical mathematics from constructing ideal continua, points without extent, exact equalities or completed infinities within its own licensed basin. It instead governs the crossing: whenever an endogenous construction is applied to the registered world, its claim must return through finite representation, uncertainty, provenance and declared conditions of admissibility.

The theory is developed through finite extent, the registration boundary, Alphonic distinction, Functional Symbolic Trajectories, symbolic drag, licensing, equality and interactional correspondence. Symbolic compression and decompression are then developed through a bead-chain model. A compressed bead can stand for a longer trajectory, while decompression reconstructs a new chain through the reader's or system's existing endogenous lattice. A scientific object becomes extant not merely because it has been named, but because a stable compression can be repeatedly reconstructed from accepted registrations and bridge operations. Part I closes with a provisional formal architecture and a set of working principles intended to support a wider second volume on language, mathematics, instruments, images, statistics, artificial intelligence and the formation of scientific objects.

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Notation and reading conventions

The notation in this first volume is deliberately light. It is intended to hold distinctions open, not to create the appearance of a completed formal system. A symbol acquires its meaning from its declared use in the trajectory of the argument. The following signs recur throughout the work.

Symbol	Working meaning
E	A physical interactional trajectory prior to symbolic identification.
G	A generonic or registration operation through which a distinction becomes symbolically available.
s_0	A first finite symbolic registration produced at a stated boundary.
s_n	A later symbolic state reached through a finite trajectory of transformations.
F_H	A Functional Symbolic Trajectory carrying an accessible history H .
α	An Alphonic distinction condition: the operative limit at which a relevant difference is registered as non-single.
Δ	A finite interactional or measurement window.
δ	The uncertainty carried by a registration or symbolic operation.
H	Accessible provenance or history.
C	Declared conditions of admissibility, consensus or licence.
L_C	A licensing operation performed under conditions C .
b_i^X	A bead instantiated within the symbolic lattice from an exogenously tethered registration.
b_i^N	A bead generated through predominantly endogenous symbolic operations.
Λ_t	The current endogenous lattice, functioning as the history-dependent handler at stage t .
$\kappa_{\Lambda,C}$	A compression operation that licenses one bead to represent a longer chain.
$\mathcal{D}_{\Lambda,C}$	A decompression operation that reconstructs a chain through a current lattice and licence.
$M \mid_C R$	A conditioned interactional correspondence between a model M and a registration R ; it does not declare identity.

Where the ordinary equals sign is used in elementary algebra, it retains its usual local mathematical function. Where the relation between a mathematical construction and a physical registration is at issue, the interactional bar is used to prevent correspondence from silently becoming identity.

Prologue: It Exists

In September 2026 a large volume arrived through the post. It had travelled from another seller, another place and, more remarkably, from the year 1727. Its title page announced Robert Greene's *The Principles of the Philosophy of the Expansive and Contractive Forces*. Beneath that declaration stood the qualification that the work concerned the several chief rational sciences which were then *extant*. At the bottom of the page were the visible supports of its first entry into the world: a named author and college, the Cambridge University Press, Cornelius Crownfield as printer, booksellers in Cambridge and London, and a Roman date impressed into paper.

The volume can be described bibliographically. Its words can be photographed, transcribed and passed into optical character recognition. A digital copy can make Greene's printed sequence available to readers who will never encounter the surviving book. In this sense the symbolic work can be propagated across many carriers. It can leave the page, enter a screen, be spoken aloud, be quoted in another book or be given to a language model as a sequence of characters.

Yet the volume on the desk is not interchangeable with those reproductions. It is not merely an instance of a title. Its paper has changed colour. Marks have been added by hands that are now unknown. Edges have been touched, bindings stressed and surfaces stained. Most of the events in its history cannot be recovered, but their loss does not mean that no trajectory occurred. The book's present state carries registrations of a history whose full sequence has disappeared.

We therefore encounter two related but non-identical things. There is Greene's symbolic construction, which can be reproduced with varying fidelity, and there is this finite physical carrier, whose particular trajectory cannot be duplicated. A photograph can transmit the appearance of a stain. It does not transmit the event that produced the stain. A transcription can preserve the letters of a sentence while abandoning the spacing, impression, texture and later marginal marks through which this copy has become what it is.

This distinction is not a nostalgic preference for an old book over a digital file. The digital file is also a finite physical and symbolic carrier. It has a format, a storage history, a sequence of transformations and conditions under which it can be read. The point is that no representation travels without a carrier, and no carrier arrives without a trajectory. A representation may conceal that trajectory, compress it or render most of it inaccessible, but it cannot have entered the present without one.

The word *extant* therefore offers a useful doorway. To call something extant is to say that it remains available to be encountered. Greene's work is extant because particular carriers

survived and because later communities continued to recognise, catalogue and reproduce its symbolic pattern. The volume is not simply present. It has been made present by an extended chain of material preservation and symbolic admissibility.

Scientific objects also become extant. A mark becomes a measurement; a set of values becomes a distribution; a pattern becomes an object; an object becomes a member of a class; a class becomes a feature of a theory. At the end of that trajectory we may speak as though the named object had stood ready-made before the observer. The chain through which it became available is compressed into the noun.

This monograph asks how that compression occurs. It begins neither with the claim that representation is illusion nor with the claim that science simply copies an independently catalogued world. It begins with the more modest observation that whatever is said scientifically must become distinguishable, registered, carried and reconstructed. Between an interaction and a sentence about the Universe lies a finite trajectory. The task is to recover the shape of that trajectory and to ask what licenses its symbolic endpoint to refer back across the boundary from which it began.

The old book has not completed its journey. A symbolic trajectory printed in Cambridge in 1727 has entered a twenty-first-century inquiry into the means by which symbols acquire and retain authority. Its continuing existence is not a metaphor for finite representation. It is an example of it.

Opening proposition

A representation is never simply present. It becomes extant through a finite carrier, a history of interactions and a community capable of reconstructing its distinctions.

The Forgotten Middle

1.1 From world to statement

Scientific prose regularly places the world and our knowledge of it next to one another. A telescope observes a galaxy. A detector finds a particle. A laboratory measures a constant. A mathematical model describes the Universe. These compact expressions are useful, but their usefulness depends upon a large compression. They remove the middle of the operation.

No instrument begins with a galaxy, a particle or a constant already inside it. Something interacts with the apparatus. Some part of the apparatus changes state. That change must persist or propagate sufficiently to become distinguishable. It is assigned a place in an instrumental scheme, converted into a mark or value, corrected, compared, grouped and interpreted. Only after this extended process does a scientific noun emerge as though it had been present at the beginning.

The elementary chain can be written as

$$\begin{aligned}
 \text{physical interaction} &\longrightarrow \text{instrumental change} \\
 &\longrightarrow \text{registration} \\
 &\longrightarrow \text{finite symbol} \\
 &\longrightarrow \text{transformation} \\
 &\longrightarrow \text{classification} \\
 &\longrightarrow \text{model} \\
 &\longrightarrow \text{scientific statement.}
 \end{aligned} \tag{1.1}$$

Every arrow in Equation (1.1) performs work. The arrows are not empty conduits through which an unchanged object passes. At each stage some distinctions are amplified and others are lost. Conditions are imposed, comparisons are selected, and a new symbolic state becomes available for the next operation. The chain is productive: it makes a scientific object speakable.

Conventional exposition often begins after most of this work has already been done. A table of values appears in a paper as “the data”. A catalogue entry appears as “the source”. A parameter obtained through a model appears as a property of an object. The paper need not repeat the entire engineering and institutional history of each symbol, but compression can create the impression that the final noun was directly observed.

The Theory of Finite Representation begins by restoring the middle. This is not a demand that every scientific sentence be replaced by an exhaustive history. Such an exhaustive history is unavailable. It is a demand that the existence of a trajectory remain part of the claim. What is not carried in the final symbol has not thereby ceased to matter.

1.2 Registration and inference

Three questions are often compressed into one:

1. What changed within the physical interaction and apparatus?
2. What was registered from that change?
3. What was inferred, named or modelled from the registration?

The first concerns a physical trajectory. The second concerns the formation of a stable distinction. The third concerns an endogenous symbolic trajectory. The questions are connected, but an answer to one cannot be silently substituted for an answer to another.

Suppose a detector supplies an ordered collection of finite values. The values may be calibrated, spatially arranged and displayed as an image. A human reader sees a bounded form and calls it an object. A processing system may perform a similar segmentation according to declared thresholds. The object is not arbitrary: its construction is tethered to registrations. Neither, however, was the noun delivered intact to the detector. The noun is an achievement of the trajectory.

This does not make the resulting science unreal. It changes the account of its realism. A scientific object is not made secure by pretending that symbolic construction never happened. Its security lies in the strength, recurrence and declared conditions of the pathway that connects registrations to the object. When different instruments, transformations and observers reconstruct a sufficiently stable relation, the bridge gains authority. That authority is historical and operational rather than magical.

1.3 Compression and symbolic drag

Scientific work must compress. A researcher cannot carry every instrumental state, calibration record, line of software, mathematical manipulation and institutional decision into each sentence. A symbol is useful precisely because it permits a large trajectory to be gathered into a manageable form. The danger appears when successful compression is mistaken for the absence of a trajectory.

The term *symbolic drag* names the residual dependence of a compressed symbol upon the path that formed it. A quantity moved into a new equation does not arrive free of its measurement conditions. A category moved into a new population does not arrive free of its selection conditions. A word moved between disciplines does not arrive with an unchanged field of meaning. The compressed symbol pulls parts of its former trajectory behind it, even when those parts are no longer visible.

Symbolic drag need not be treated as an error term that can always be removed. It is the structural consequence of finite representation. Some history is retained explicitly; some survives indirectly in uncertainty, conventions or calibration; some is lost. A theory of representation must therefore ask not only whether a transformation can be performed, but

also what the new symbol continues to carry and what its compression has abandoned.

The forgotten middle

The distance between physical interaction and scientific statement is not a blank. It is a finite productive trajectory in which distinctions are created, transformed, compressed and licensed.

Three Basins

2.1 Why separation is required

The Theory of Finite Representation concerns relations between physics, mathematics and language, but it must not begin by mixing them. If the properties of one domain are silently transferred to another, the bridge has already been crossed before its conditions can be examined. A clearer architecture separates three co-related basins: the physical basin, the symbolic basin and the bridge basin.

The word *basin* is used because each domain supports families of trajectories that can stabilise under its own commitments. It is not intended to impose a rigid container upon every event. A physical registration may enter a symbolic process, and a symbolic instruction may alter the arrangement of an instrument. What matters is that the direction and conditions of the crossing are declared.

2.2 The physical basin

The physical basin contains interactional trajectories, changes of state, apparatus, carriers and the finite extents across which registrations are formed. It does not initially contain ready-made equations or named objects. The phrase “does not initially contain” is important. The apparatus may have been constructed under a mathematical design and used under linguistic instructions, but its physical change is not itself a numeral or sentence.

At this level the first admissible statement is deliberately spare: something interacted, and some difference became physically available to a registration system. We may later identify the source, mechanism and quantity. Those are subsequent achievements. The restraint prevents a mature symbolic description from being projected backwards and treated as though it were the unmediated content of the event.

The physical basin is therefore not a claim about a world wholly independent of every observer. It is the working domain in which interaction, extent and change precede their symbolic classification. It marks the exogenous side of the registration boundary.

2.3 The symbolic basin

The symbolic basin contains finite marks and the operations performed upon them. Words, numerals, equations, diagrams, tables, classifications, proofs, programs and narratives can all unfold within it. Once a registration has entered this basin, it can participate in transformations whose immediate constraints are endogenous to the declared system.

Mathematics is exceptionally powerful at constructing and preserving such endogenous trajectories. Given accepted axioms and rules, a mathematician can move from one symbolic state to another without returning to an instrument at each step. An ideal point, a continuum or an infinite set can be admissible within this basin because the licence governing the construction permits it. The Theory of Finite Representation does not need to enter the mathematical basin and confiscate those objects.

Language likewise supports long endogenous trajectories. A reader can build a landscape, an institution or a flying creature from finite sentences. The imagined object acquires weight, colour and behaviour as constraints accumulate. The trajectory is real as a symbolic and cognitive process even when it is not tethered to an exogenous physical registration of the object it describes.

2.4 The bridge basin

The bridge basin contains the operations by which a symbolic construction is related to a physical registration. Calibration, operational definition, measurement protocol, model fitting, prediction and experimental comparison belong here. So do the institutional and linguistic acts through which a community declares that a particular correspondence is admissible.

The bridge is not a narrow final step added after physics and mathematics are complete. It operates at many levels. An instrument design maps a mathematical expectation into a physical arrangement. A calibration maps a physical response into a numerical scheme. A model maps symbolic relations into expected registrations. An experimental comparison maps new registrations back into a judgement about the model.

The architecture can be written as

$$\boxed{\text{physical interactional trajectory}} \xleftrightarrow{\text{declared bridge rules}} \boxed{\text{functional symbolic trajectory}}. \quad (2.1)$$

The arrow is double because scientific practice crosses in both directions. It is not an equals sign. The two trajectories can correspond without becoming one thing.

2.5 Correspondence without miniature worlds

A mathematical model is often spoken of as though it were a simplified copy of reality. That picture invites the question of whether the model is true in the sense that each mathematical object has a matching physical object. The three-basin architecture suggests another account. A model is a symbolic dynamical structure whose finite outputs may correspond, under

stated bridge rules, to registered physical trajectories.

The model need not resemble the world in the manner of a scale object. It must produce relations that can be connected to possible registrations. Its utility may depend upon how reliably those relations survive changes of instrument, context and representation. The bridge can be narrow or broad, fragile or robust, local or repeatedly extended. None of those properties is established by the internal elegance of the model alone.

This separation permits a constructive pluralism. Alternative mathematical trajectories can be explored without first declaring that one alone constitutes the ontology of the physical world. Different models may preserve different registered relations. The Theory of Finite Representation asks how their bridges are constructed and what each bridge licenses us to say.

Three-basin rule

No property declared inside one basin crosses into another without a bridge operation. Correspondence may be strong, repeatable and productive without being converted into identity.

Finite Extent

3.1 Registration takes place

To register is to undergo and preserve a difference. Preservation need not be permanent, but it must endure sufficiently for a later state or system to be conditioned by it. A durationless event that leaves no distinguishable change cannot enter a scientific trajectory. Registration therefore occupies an extent.

This extent may involve time, spatial separation, energy transfer, material change or another ordered relation. The theory does not begin by assuming that all extents can be reduced to one master coordinate. It begins with the local fact that a relevant distinction requires a finite window within which it can form.

Let Δ denote such an interactional window. The statement

$$\Delta > 0 \tag{3.1}$$

does not establish an absolute smallest interval. It states that the particular registration did not occur across a literally absent extent. A camera exposure, an electrical conversion, a mark made on paper and a spoken syllable have very different mechanisms, yet none is available as a registration without an extended change.

The word *instantaneous* often contracts this requirement out of the description. It may be a useful limiting instruction inside mathematics, but the physical registration still has to begin, stabilise and become available. The theoretical instant and the measurement window belong to different basins. Their connection requires a bridge.

3.2 Measurable and minimum measurable extent

Three statements should be separated. First, an actual measurement has a finite extent. Second, under stated conditions there may be a minimum extent at which a chosen apparatus can distinguish the relevant change. Third, there may or may not be a universal lower physical extent. The first two can be investigated without settling the third.

For an apparatus A operating under conditions C , write

$$\Delta_{\min}(A, C) \tag{3.2}$$

for the smallest interactional window within which a chosen distinction can be repeatedly registered. This is not an immutable property of nature. Change the apparatus, the criterion or the environment, and the value may change. The notation records the conditions rather than hiding them.

The same discipline applies to spatial distinction. An instrument does not measure a Euclidean point. It forms a response over a finite sensitive region and a finite interaction. Mathematical localisation can become arbitrarily fine inside a model; physical localisation returns through an apparatus whose response has extent.

This is why “finite” must not be equated with crude. Finite registrations can be extraordinarily discriminating. Their precision is an achievement of stabilised interaction, engineering, repeated comparison and symbolic control. The theory does not diminish precision. It gives precision a trajectory.

3.3 Extent before object

Ordinary language encourages us to begin with things. An object exists, and then we measure one of its properties. At the registration boundary the order may be reversed. A finite difference is registered; relations among registrations are stabilised; only then does a symbolic object become available.

This reversal matters most where the middle-scale noun is least secure. At ordinary human scales, a chair or a tree can be repeatedly encountered from many positions, and language has long stabilised the corresponding nouns. At very small and very large scales, the relation between registration and noun is much longer. A “particle”, “field”, “galaxy”, “void” or “expansion” enters discourse through extensive instrumentally mediated trajectories.

The theory does not infer that such objects are fictitious. It asks us to avoid placing the completed noun at the start of its own construction. Extent and interaction precede the licensed object in the order of registration.

3.4 The restrained physical claim

The initial physical commitment of this monograph can now be stated.

Principle of Finite Extent

Whatever the unregistered physical world may be, anything entering an empirical scientific trajectory must produce a finitely distinguishable change across a non-zero interactional extent.

This principle does not declare that nature is composed of indivisible finite units. It does not establish a universal atom of space or time. It places a condition upon scientific access. A claimed physical entity may be modelled by an ideal continuum, but the evidence supporting that claim must still become available through finite registration.

This restrained formulation is important. A theory of finite representation should

not smuggle an entire ontology into its first axiom. Its task is to follow the chain from interaction to representation and back. More ambitious claims may later be proposed and experimentally examined, but the architecture does not depend upon their prior acceptance.

The Registration Boundary

4.1 From interaction to distinction

An interaction does not become a scientific registration merely because something happened. The interaction must alter a carrier in a way that can be distinguished by a subsequent operation. The boundary at which this occurs is called here the *registration boundary*. In earlier work it has also been called the generonic boundary because it is the region in which a usable symbolic distinction is generated.

The boundary is not necessarily a single physical surface. In a detector it may be distributed across sensing material, charge accumulation, amplification, analogue conversion, thresholding and storage. In reading it may extend across light, page, eye, neural processing and learned recognition. The boundary is functional: it identifies the crossing at which a change becomes available as a member of a symbolic trajectory.

The elementary operation is written

$$E \xrightarrow{G_{\alpha,\Delta,\delta}} s_0, \quad (4.1)$$

where E is a physical interactional trajectory, G is the registration operation, α states the operative distinction condition, Δ is the finite interactional window, δ records uncertainty and s_0 is the first finite symbol admitted to the subsequent trajectory.

Equation (4.1) should not be read as a claim that the entire physical event has been converted into a complete symbol. The symbol is a conditioned registration produced through the interaction. It preserves some relations and abandons others. The arrow marks an irreversible historical fact: after the registration, the symbolic trajectory may continue even though the original interaction is no longer available.

4.2 Analogue continuity and symbolic decision

An apparatus may respond continuously over some range, but the production of a usable registration requires a decision that a difference counts. Even an apparently continuous trace is read through finite width, sampling, display, resolution and comparison. The theory does not need to deny continuity in the pre-symbolic response. It asks how a distinction is drawn from it.

This provides a more careful role for analogue-to-digital conversion. An ADC is a

conspicuous generonic mechanism because it assigns bounded physical responses to finite codes, but digitisation does not create the entire problem. A scale mark, photographic grain, spoken phoneme and hand-drawn boundary also stabilise differences under finite conditions. The digital device makes the decision architecture unusually visible.

At the registration boundary, a range of possible interactions is compressed into an admitted state. If the device yields k , it has not copied an infinitely exact magnitude into a numeral. It has placed the event within an operational distinction scheme. A fuller registration should therefore be thought of not simply as k , but as

$$\mathcal{R}_0 = (k, \alpha, \Delta, \delta, H, C), \quad (4.2)$$

where the added terms record the conditions that make k usable. Ordinary practice may store only the first component and parts of the uncertainty. The remaining components do not cease to exist because they were not carried forward.

4.3 Registration is not passive copying

The metaphor of an instrument as a window suggests that the world passes through unchanged and becomes visible on the other side. Instruments are more active than windows. Their materials, geometry, thresholds and timings select which interactions can produce stable changes. Their processing systems decide which changes will be retained. Their output formats determine which relations are available to subsequent symbolic operations.

This participation does not make measurement arbitrary. A key must fit a lock before the lock can turn, but the lock does not invent the key. In a similar way, an apparatus can constrain and transform an interaction without becoming its sole cause. The registered state is relational: it is formed by the encounter between an interactional trajectory and a system capable of being changed by it.

The relation also prevents us from imagining a measurement as the discovery of a symbol already written in nature. Numbers are not physically delivered to an ADC. The device produces a code according to its construction and declared scale. The code can be extraordinarily reliable and physically tethered while remaining a representation.

4.4 The first symbol is already historical

It may seem that provenance begins only after data have been stored and processed. Yet s_0 already has a history. The apparatus was made and calibrated. A measurement window was chosen. Thresholds were set. Environmental conditions shaped the response. The source and detector occupied a particular configuration. The first symbol is therefore not an ahistorical seed from which history later grows.

Nor can that history be wholly recovered. The present registration does not contain a complete replay of the interaction that produced it. Provenance is necessarily partial. Its role is not to offer impossible omniscience but to preserve enough of the trajectory for the claim to be examined, repeated, reconstructed or challenged.

Registration boundary

A registration is a finite relational event in which physical interaction produces a sufficiently stable distinction to condition a later symbolic state. The resulting symbol is tethered to, but not identical with, the event.

The Axiom of Finite Representation

5.1 The instantiated symbol

Mathematics can refer to a symbol without specifying whether it is written in ink, displayed by pixels, held in memory or spoken aloud. This abstraction is indispensable. It permits the same mathematical relation to be reconstructed across different carriers. Yet every occasion on which the relation is used requires an instantiation.

An instantiated symbol occupies a finite medium. It has a form sufficiently distinct from neighbouring forms to be recognised. It is created and preserved across a finite interval. It has a bounded recoverable history. Even when its content refers to infinity, the present mark does not become infinite. The finite inscription ∞ can license operations within a formal system, but the inscription and the licensed mathematical object are not the same category of thing.

This distinction allows the proposed axiom to be stated without attempting to rewrite all of mathematics.

Axiom of Finite Representation

Every instantiated representation has finite extent, finite distinguishability, bounded resolution and a physical or historical provenance.

The axiom concerns representation as performed and carried. It does not forbid a formal system from referring to ideal or infinite objects. It asserts that every available expression of that reference is finite and that any return to physical comparison must occur through another finite registration.

5.2 A scientific strengthening

For empirical science, the axiom can be strengthened:

Scientific form

Any symbol admitted into an empirical scientific trajectory must be finitely instantiated and must arise through, or be traceable to, a finite chain of registrations and declared symbolic transformations.

The phrase “traceable to” does not require complete historical recovery. A centuries-old observation may have incomplete documentation; a processed dataset may have lost intermediate states. The incompleteness itself becomes part of the admissibility judgement. A symbol with missing provenance can still be used, but it cannot silently acquire the same status as a symbol whose chain is preserved and independently reconstructed.

Nor does a long symbolic derivation require a fresh physical measurement at each step. Once admitted, symbols can participate in endogenous mathematics. Traceability requires that the initial conditions and later bridge claims be declared. It does not reduce mathematics to a list of laboratory operations.

5.3 Six senses of the finite

The word *finite* carries several connected meanings within the axiom. Separating them prevents one meaning from being substituted for another.

First, the carrier has finite physical extent. A page, memory device, acoustic wave or neural state does not occupy an absent region. Second, the inscription is finite: only a bounded sequence of marks is present in any instance. Third, the distinction is finite: the mark is recognised through some operative resolution. Fourth, the precision is finite: an empirical value is registered within an uncertainty structure. Fifth, the operational trajectory is finite: the present representation was reached through a bounded sequence of available operations. Sixth, the recoverable provenance is finite: only part of the symbol’s history remains accessible.

These six senses are related but not identical. A short formula may refer to a mathematical construction whose internal definition is not finitely enumerable. A physical carrier may be large while its symbolic alphabet is small. A highly precise value may have a brief inscription but an enormous experimental history. The theory must therefore state which finitude is doing work in each argument.

5.4 Numbers as measurements

Within pure mathematics a numeral participates in rules determined by the formal basin. Within measurement a number is a registered symbolic value. It is not best described as an imperfect approximation to an infinitely exact number hidden behind the apparatus. It is the measured value produced under the stated conditions, accompanied by uncertainty and provenance.

This shift is subtle but important. If a length is reported as 4.20 units under a specified method, the measurement is not pretending to be an endless decimal that has been carelessly truncated. The reported symbol is the finite outcome of the operational trajectory. Further

measurement may alter the value, reduce an uncertainty or disclose structure within the earlier window, but it does not transform the earlier registration into a metaphysical failure.

The distinction also clarifies why computational finitude is only one part of the theory. A computer stores finite strings, but those strings already depend upon a scheme of symbol formation and interpretation. The important question is not merely how many bits are available. It is how the stored states are tethered to interactions, transformed through trajectories and licensed to refer beyond themselves.

5.5 The axiom at the crossing

Earlier versions of finite mathematics might attempt to insert the axiom into the internal foundations of every mathematical system. The present architecture assigns it a more exact role. Mathematics may establish its own commitments. The axiom governs any crossing by which a mathematical construction acquires empirical application.

An infinitely differentiable continuum can remain admissible inside its formal basin. If a physical theory declares that the world *is* such a continuum, an additional move has occurred. The mathematical licence has been extended across the bridge. The Theory of Finite Representation asks what registrations and correspondence rules license that extension.

This question cannot be answered merely by saying that the mathematics works. The work is demonstrated when finite predictions are compared with finite registrations. The comparison may strongly support the model, but it does not retroactively convert every internal mathematical commitment into an observed physical property. Application licenses correspondence, not automatic identity.

The Conditions of Representation

6.1 From pillars to a working state

Finite representation requires more than a bounded mark. A mark must be distinguishable, its uncertainty must be carried, its history must be sufficiently recoverable and its use must be admitted under some declared or inherited conditions. These requirements can be assembled as five conditions: finite symbol, Alphonic distinction, uncertainty, provenance and admissibility.

A working representational state can be written

$$\mathcal{R} = (s, \alpha, \delta, H, C), \quad (6.1)$$

where s is the present finite symbol, α the distinction condition, δ its uncertainty, H its accessible history and C the conditions under which it is admitted. Equation (6.1) is not a complete ontology of symbols. It is a reminder that the visible mark is only one component of a scientifically usable representation.

6.2 Finite symbol

A finite symbol is a bounded difference capable of participating in a later trajectory. It can be a letter, a numeral, a pixel value, a line, a sound, a gesture or a compound structure built from simpler distinctions. Its identity is not given by material sameness. The letter printed in this sentence and the corresponding letter on another screen are not the same physical object. They are admitted as instances of a symbolic class through a learned and socially stabilised correspondence.

The symbol therefore already contains a compressive operation. Variations in size, ink, position and carrier are disregarded so that a stable class can be reconstructed. The compression is productive; without it, writing could not travel. But it also means that symbolic identity is conditioned rather than absolute.

6.3 Alphonic distinction

The Alphonic condition concerns the smallest difference that counts as non-single within the present operation. The term does not refer only to alphabetic characters. It names the

boundary at which one potential state is distinguished from another and can therefore enter a symbolic trajectory.

Write $\alpha(A, C)$ for the operative distinction condition of apparatus or observer A under conditions C . Two physical variations that fall within the same admitted state are symbolically single for that operation, even if a different apparatus could separate them. Conversely, a difference larger than the threshold may still fail to register if it is not coupled to the chosen system.

The search for a universal physical Alphonic limit is an important experimental question, but it is not required for the local theory. The theory begins with measurable distinction conditions. It can later ask whether those conditions converge upon a common interactional structure.

6.4 Uncertainty

Uncertainty is often appended to a value after the value has been declared. In finite representation it belongs to the formation of the distinction. The registration is produced across a response window, under finite resolution, through interactions that cannot be wholly reconstructed. Its uncertainty is therefore part of what the symbol is licensed to do.

The symbol s without δ is a stronger compression than the scientific state warrants. It may be convenient in subsequent calculation, but the trajectory should retain a route back to the uncertainty that was suppressed. When uncertainty is propagated, transformed or replaced by a statistical summary, those operations become part of the provenance.

Uncertainty need not be interpreted as ignorance of a perfectly exact hidden value. It can describe the finite span over which the present distinction is stabilised. This permits measured values to be treated as genuine finite registrations rather than defective infinities.

6.5 Provenance

Provenance H is the accessible history by which a representation became present. It includes physical production, instrumental settings, transformations, selection, storage, transcription and interpretive commitments. No scientific symbol carries all of this within its visible form. Provenance is distributed across records, practices, institutions and other symbols.

Complete provenance is impossible. Even the old Greene volume does not tell us who touched each page or when each stain occurred. The requirement is not total history but honest bounded history. A claim should not gain strength by forgetting that parts of its chain have been lost.

Provenance also distinguishes two symbolically similar objects. Two files may contain identical sequences of characters while having different sources and uses. Two measurements may share a numerical value while arising from different instruments. Treating them as identical can be legitimate for one operation and inadmissible for another. The required level of provenance is therefore conditioned by the claim.

6.6 Admissibility and consensus

A private mark becomes scientifically effective only when it can enter a shared trajectory. Admissibility C gathers the rules, practices and communal judgements under which a representation is accepted for a particular purpose. Some conditions are explicit, as in a calibration standard or mathematical axiom. Others are inherited through training, disciplinary language and institutional practice.

Consensus is not a guarantee that a representation is physically correct. It is the stabilising mechanism through which symbols become usable across a community. A standard unit, catalogue class or proof convention permits many people to reconstruct sufficiently similar operations. The resulting authority can be examined historically without being dismissed as arbitrary.

Admissibility is therefore both enabling and limiting. A licence opens a basin of possible work while excluding trajectories that do not satisfy its conditions. The Theory of Finite Representation makes that licence visible so that alternative commitments can be compared rather than treated as unthinkable.

Representational state

The usable scientific symbol is not s alone but the conditioned state $\mathcal{R} = (s, \alpha, \delta, H, C)$. Compression may hide the other components; it does not abolish them.

Functional Symbolic Trajectories

7.1 Symbols in motion

A symbol is functional because it alters what can happen next. A mark in a calculation licenses an operation; a word changes the reader's developing landscape; a measurement constrains a model; a classification directs an object into a population. Meaning is therefore not exhausted by a static dictionary relation. It is carried by the symbol's position and effect within a trajectory.

A Functional Symbolic Trajectory is written

$$s_0 \xrightarrow{f_1} s_1 \xrightarrow{f_2} s_2 \xrightarrow{f_3} \dots \xrightarrow{f_n} s_n, \quad (7.1)$$

where each f_i is an admissible transformation under some conditions C_i . The states need not be single characters or numbers. A state may be a sentence, diagram, table, matrix, proof position, instrumental frame or composite document. What makes the sequence an FST is that the transitions carry constraints forward and alter the field of available continuations.

The linear writing of Equation (7.1) should not be mistaken for a claim that the underlying process has only one dimension. A trajectory can branch, revisit earlier structures, draw upon memory and interact with other trajectories. The written line is itself a finite projection that allows the process to be inspected.

7.2 Unfolding the compressed object

A scientific object often appears at the end of a long FST:

$$\text{detector change} \longrightarrow \text{value} \longrightarrow \text{corrected value} \longrightarrow \text{feature} \longrightarrow \text{class} \longrightarrow \text{object}. \quad (7.2)$$

Once the object is stable, later work may begin with it as though it were a primitive. That is often efficient. The Theory of Finite Representation adds a complementary operation: unfold the noun and recover as much as possible of the trajectory through which it became extant.

Unfolding is not a demand to abolish objects. It is a method for examining their bridges. Where did the exogenous registrations enter? Which operations were endogenous? Which

distinctions were amplified? Which variations were compressed into one class? What uncertainty and history survived into the endpoint? These questions turn ontology back into an inspectable trajectory. Chapter 10 will give this unfolding a more exact operational name: decompression through a current endogenous handler.

7.3 The reader becomes an author

An FST does not move intact from writer to reader. The reader receives a finite sequence of marks and reconstructs relations through a previously formed linguistic and experiential basin. Reading is therefore a new trajectory constrained by, but not identical with, the written one.

This can be seen when a fictional object is built gradually. Begin with the phrase “a flying pig”. Add that it is a piglet small enough to sit in one hand, that it weighs about as much as a tennis ball and that its wings are thin and webbed. No physical pig has entered the room, yet writer and reader can develop increasingly constrained and partially corresponding objects. Each new sentence reshapes the available imaginative trajectory.

The example matters because scientific reading also requires reconstruction. A displayed astronomical image, a graph or an equation does not force one complete interpretation into the observer. Learned distinctions, expectations and surrounding language guide the reconstruction. The difference between the flying pig and the scientific object lies not in the absence of symbolic construction from one of them. It lies chiefly in provenance: the scientific trajectory is claimed to be tethered, through declared bridge operations, to exogenous registrations.

7.4 Trajectory loss and statistics

When many registrations are gathered into a distribution, much of their individual history is compressed. The statistical structure can disclose relations that are not visible in single events, but the process also changes what is available. Order may be discarded, local dependence removed and different paths rendered exchangeable.

The resulting loss is often described through entropy. From the present perspective, an important part of the phenomenon is trajectory loss: states remain as values while the longer paths that produced them are no longer recoverable. Two datasets with similar distributions can have arisen through different dynamics. A theory restricted to the endpoints may treat them as the same; an embedded trajectory may reveal different attractors or transitions.

This creates a later programme for Finite Symbolic Dynamics. An equation, proof, sequence of measurements or corpus can be unfolded step by step and embedded as a time-extended symbolic trajectory. Poincaré sections and reconstructed attractors then become possible analytic tools rather than metaphors. Part I establishes why such an analysis is required; later work must develop its practical methods.

7.5 Symbolic drag revisited

The endpoint s_n carries constraints inherited from earlier states even when their explicit form has been removed. This is symbolic drag. In language it appears when a word brings historical and disciplinary associations into a new sentence. In mathematics it appears when an abstract object retains the commitments of its construction. In measurement it appears when a number is reused without its instrumental conditions.

Symbolic drag is not uniform. Some prior constraints fade; others dominate the new basin. A stable noun can conceal an enormous trajectory while exerting a strong pull upon every sentence in which it appears. One purpose of unfolding is to identify that pull and to decide whether the inherited commitment remains admissible.

Trajectory principle

A finite symbol is a temporary stabilisation within a history of transformations. Its function and admissibility depend upon the trajectory that brought it into the present and the continuations it makes available.

The Bridge Axiom

8.1 The missing rule of application

Mathematics can proceed endogenously from declared axioms and rules. Physics depends upon registered interactions. Their extraordinary partnership is often treated as self-explanatory: mathematics works, therefore the physical world must in some sense be mathematical. But the success of a correspondence does not explain the operation by which a symbolic construction was licensed to acquire physical reference.

The missing rule is not another equation within a physical model. It is a rule governing the model's application. It asks which finite registrations are connected to which mathematical states, under what transformations and within what limits. The proposed bridge axiom is therefore stated as follows.

Bridge Axiom

No endogenous mathematical or linguistic construction acquires physical identity solely from its internal coherence. Its empirical application requires a declared correspondence with finite registrations under stated conditions of distinction, uncertainty and provenance.

The phrase “solely from” leaves room for mathematical coherence to matter. An inconsistent model cannot provide a stable bridge. Coherence is necessary for many applications, but it is not sufficient to turn a formal object into a physical object. The crossing requires an additional trajectory.

8.2 The inward crossing

In the direction from physical interaction towards mathematics, the crossing has the form

$$E \xrightarrow{G_{n,\Delta,\delta}} s_0 \xrightarrow{F_H} s_n \xrightarrow{L_C} M, \quad (8.1)$$

where M is a mathematical or linguistic construction licensed from the symbolic trajectory. The notation does not imply that M contains the event E . It marks the sequence through which relations registered from E came to constrain M .

The inward crossing is often presented as data entering a model. Yet “data” is already a late symbolic state. Before a value becomes data, a registration system must decide that it belongs to an admissible kind, attach units or labels, place it within an order and preserve it in a form available for processing. The bridge begins before the model receives its table.

The same model can receive values from different instruments because bridge rules establish operational correspondences among their registrations. This transfer is a scientific achievement. It should not be mistaken for proof that the physical mechanisms or complete trajectories of the instruments are identical.

8.3 The outward crossing

In the reverse direction a model produces an expected finite representation:

$$M \xrightarrow{P_C} \hat{\mathcal{R}} \mid_C \mathcal{R}_{\text{new}}, \quad (8.2)$$

where P_C is a prediction operation, $\hat{\mathcal{R}}$ is the expected registered state and $\mathcal{R}_{\text{new}}$ is a new physical registration. The interactional bar records a conditioned comparison. A prediction is not compared with the unregistered world in itself. It is compared with a state produced through another generonic boundary.

This is why prediction always returns to finite form. A theory may be defined over a continuum and use exact internal relations, but an empirical test asks whether some finite distinction appears within a measurement window. The success of that test supports the bridge over the examined region. It need not license every mathematical feature of the model as a physical identity.

Repeated successful crossings can make the bridge extremely robust. They can support engineering, navigation, medicine and new instruments. The theory of finite representation does not diminish this success. It describes more carefully what has succeeded: a network of conditioned correspondences between symbolic trajectories and reproducible registrations.

8.4 Bridge strength

Not every bridge has the same authority. A useful theory will eventually need ways to describe bridge strength. Several dimensions are already visible.

A bridge may be recurrent if the correspondence is repeatedly reconstructed. It may be independent if different apparatuses and groups recover related registrations. It may be transportable if the relation survives a change of context. It may be discriminating if nearby alternative models produce different expected registrations. It may be provenance-rich if the production and transformation of the symbols are preserved. It may be narrow if it works only within a small operational domain.

These are not reducible at once to a single score. A bridge can be precise but narrow, broad but weakly discriminating, or recurrent within one institutional pipeline while poorly reproduced elsewhere. The language of bridge strength encourages a plural and inspectable account of scientific confidence.

8.5 Alternative trajectories

Once correspondence is separated from identity, alternative symbolic trajectories become available. Two models may map different internal objects onto the same finite registrations. Within the tested window they may be empirically indistinguishable. The conventional response may be to choose the more elegant or established model. The Theory of Finite Representation records that choice as an additional licensing decision rather than as a measurement.

This does not force permanent indecision. New instruments can generate new distinctions that separate the models. A model may prove more generative by opening bridges to previously unrelated registrations. But until that occurs, the unregistered difference between the models remains endogenous.

Geofinitism can therefore be understood as a chosen exploratory trajectory. It begins with finite extent and finite distinction, then asks what mathematics, physics and language can be built while preserving those commitments. Its value need not depend upon the immediate destruction of every classical path. It offers another route through the landscape and allows the resulting bridges to be compared.

Equality, Identity and Licence

9.1 The most compressive sign

The equals sign is one of the most powerful operations in symbolic culture. It allows different inscriptions to be treated as one value, different expressions to be substituted, and long trajectories to terminate in a compact declaration. Its power lies in compression.

Within a formal basin, the compression is governed by declared rules. If $A = B$ has been established, the system licences substitutions that would not otherwise be available. The relation is productive because distinctions judged irrelevant to the operation are removed.

Difficulty arises when formal equality crosses into measurement without a change of status. Two measured values have different physical histories even when their visible numerals agree. Repeated measurements occupy different interactional windows. Different instruments respond through different mechanisms. Their equivalence is conditioned by a tolerance, a unit system, a calibration history and the purpose of the comparison.

The statement

$$x_1 = x_2 \tag{9.1}$$

can therefore hide a longer empirical claim:

$$x_1 \mid_C x_2, \tag{9.2}$$

meaning that the registered values are admitted as corresponding under conditions C . The bar does not make the comparison vague. It requires the conditions of precision to remain visible.

9.2 Formal equality

Formal equality belongs to a symbolic system whose objects and rules have already been licensed. It can be exact because exactness is part of the game's construction. The finite physical inscription of the proof need not resemble the mathematical objects to which it refers. What matters internally is that the permitted symbolic transformations preserve the declared relation.

The Theory of Finite Representation does not need to replace every mathematical equals sign with an uncertainty interval. That would confuse the symbolic and physical basins. It asks instead where the expressions originated, what operations they license and whether a later physical interpretation preserves the distinction between formal identity and measured correspondence.

This separation also permits mathematics to be studied dynamically. A proof can be unfolded as an FST whose states and permitted transitions are explicit. Equality then appears as an operation that identifies nodes or trajectories under the proof's licence. The proof does not leave the symbolic basin merely because its conclusion may later be useful in physics.

9.3 Measured equivalence

Measured equivalence is established within a resolution. Suppose two registrations are given by

$$\mathcal{R}_1 = (s_1, \alpha_1, \delta_1, H_1, C_1), \quad \mathcal{R}_2 = (s_2, \alpha_2, \delta_2, H_2, C_2). \quad (9.3)$$

Their visible values may match while their distinction conditions, uncertainties and histories differ. To declare them equivalent is to select a comparison operation C^* under which those differences do not alter the present task:

$$\mathcal{R}_1 \mid_{C^*} \mathcal{R}_2. \quad (9.4)$$

Another task may require the differences to be restored. A clinical instrument and a research instrument may produce readings that are equivalent for one decision but not for another. Equality detached from use cannot express this conditional structure.

Measured equivalence is thus not a weakened formal equality. It is a different kind of relation. It compares finite trajectories rather than collapsing them into a timeless identity.

9.4 The declaration hidden in the sign

Every equality rests upon a licence to disregard difference. In elementary arithmetic that licence is so stable that it disappears from attention. In physical and linguistic use it can become contentious. When a theoretical parameter is identified with a property of nature, or a statistical class with a physical population, the sign may carry an undeclared narrative across the bridge.

The historical form of the equals sign is therefore less important here than its present functional role. It does not merely report sameness. It declares which future substitutions are permitted. It changes the available symbolic trajectory.

The interactional bar is introduced to keep the encounter open:

$$M \mid_C R. \quad (9.5)$$

The model and registration remain different trajectories, joined under conditions C. The relation can be refined, repeated, rejected or extended. No infinite middle is required in which their distinctions vanish.

9.5 Licence as productive constraint

A licence is not merely permission granted by an authority. It is a set of commitments that makes a basin of trajectories possible. Euclidean geometry, set theory, a measurement standard and a scientific catalogue each license some operations while excluding others. Once the licence is stabilised, participants can work rapidly without restating it at every step.

The productive success of a licence can make its starting commitments appear inevitable. The Theory of Finite Representation restores them as choices with histories. This does not make every choice equally useful. It allows their consequences and bridges to be examined.

Scientific consensus performs a similar function. It compresses many competing trajectories into a shared vocabulary and permits collective work. Yet the resulting noun remains a licensed stabilisation, not an escape from history.

Equality can therefore be read as a particularly strong compression: two previously distinguishable symbolic paths are licensed to occupy one operative position. The next chapter generalises that operation from equality to the compression and decompression of finite bead-chains.

Equality and the bridge

Formal equality is licensed identity within an endogenous system. Empirical comparison is conditioned correspondence between finite trajectories. The two relations may cooperate, but they must not be silently exchanged.

Symbolic Compression and Decompression

10.1 From chain to label

The previous chapter considered equality as a licensed removal of distinction. That operation belongs to a more general symbolic mechanism. A finite chain of beads can be compressed into a further bead which is then permitted to stand for the chain. Let

$$B = (b_1, b_2, \dots, b_n) \tag{10.1}$$

be an ordered symbolic trajectory. Under conditions C , and through the present endogenous lattice Λ , compression produces a labelled bead β :

$$(b_1, b_2, \dots, b_n) \xrightarrow{\kappa_{\Lambda, C}} \beta. \tag{10.2}$$

The new bead does not contain the earlier chain as a small physical object. It is licensed to represent that chain in later operations. The chain may remain recoverable elsewhere, be preserved only in part, or disappear almost entirely from accessible provenance. What survives in β is sufficient for some continuations and insufficient for others.

Compression is therefore more specific than a general reduction of symbolic distance. It involves a representational substitution. One bead becomes a handle for many beads, allowing a longer trajectory to be moved, compared and inserted into a new chain without being repeated in full. A noun, parameter, class, equation or theory can all function in this way.

The operation is productive because finite symbolic systems could not carry every preceding distinction into every new state. A reader cannot reconstruct the full history of each word whenever it appears. A calculation cannot repeat the derivation of every quantity at every line. Compression makes continued movement possible. Its cost is that the visible bead no longer displays the trajectory for which it stands.

10.2 Exogenously tethered and endogenously generated beads

Not every bead has the same provenance. Some arise when an interaction crosses the registration boundary. Others are formed predominantly through operations already taking place inside the symbolic lattice. This difference can be marked without claiming that two different substances exist inside the lattice.

An exogenous interaction E must first produce a finite registration:

$$E \xrightarrow{G_{\alpha, \Delta, \delta}} b_i^X \in \Lambda_t. \quad (10.3)$$

The bead b_i^X is now instantiated within the endogenous lattice Λ_t , but its superscript records an exogenous tether. The external event has not entered the lattice intact. A finite internal record has been formed from an interaction with it.

An endogenous bead is formed by operating upon beads already available in the lattice:

$$(b_1^N, b_2^N, \dots, b_m^N) \xrightarrow{\Phi_{\Lambda_t, C}} b_{m+1}^N. \quad (10.4)$$

The superscripts X and N therefore identify provenance rather than material kind. Once instantiated, both kinds can occur in the same chain. They can be compared, compressed and used to generate further beads. The important difference is whether the available history includes a declared route back to an exogenous registration.

This distinction also prevents a completed physical noun from being placed at the beginning of its own measurement. What enters the lattice is not the galaxy, particle, sound or length as a ready-made symbolic object. What enters is a registered bead from which longer endogenous constructions can begin.

10.3 Decompression as reconstruction

Decompression moves in the other direction. A compressed bead is used to generate a longer chain:

$$\beta \xrightarrow{\mathcal{D}_{\Lambda_t, C}} \hat{B}_t = (\hat{b}_1, \hat{b}_2, \dots, \hat{b}_q). \quad (10.5)$$

The hat over B is essential. In an engineered lossless codec, compression and decompression may be arranged so that the recovered finite chain is identical to the original under the rules of that codec. Linguistic and scientific decompression are usually different. The compressed bead does not determine one exhaustive reconstruction. The chain is generated through the current lattice Λ_t , its accumulated history and the present licence C .

Two readers can therefore decompress the same word into related but non-identical chains:

$$\mathcal{D}_{\Lambda_a, C}(\beta) \neq \mathcal{D}_{\Lambda_b, C}(\beta). \quad (10.6)$$

Communication remains possible because language stabilises enough shared constraints for the reconstructed chains to correspond. It does not require the internal trajectories of the

readers to become identical. The shared word is a finite perturbation entering two different histories.

Decompression is consequently more than retrieval. It can add beads that were never explicitly present in the compressed carrier. A remembered scene, an inference, an emotional association or a new mathematical consequence may appear as the chain unfolds. These additions are not necessarily errors. They are evidence that a functional symbol acts through a handler rather than opening like a container.

10.4 Reading through a handler

The endogenous lattice Λ_t functions as the handler of the incoming chain. It is not a passive dictionary in which each word has one stored meaning. It is the present configuration of learned distinctions, active context, remembered trajectories, expectations and already constructed beads. Because this configuration changes during reading, the handler at stage $t + 1$ is not identical to the handler at stage t .

Reading begins when an external sequence of marks or sounds produces a chain of registered beads. That chain is coupled to the present lattice, decompressed and compared with existing chains. A new result can then be injected directly into the active trajectory:

$$E_{\text{in}} \xrightarrow{G} B^X \xrightarrow{\mathcal{D}_{\Lambda_t, C}} \widehat{B}_t \xrightarrow{\Psi_{\Lambda_t}} r^N \longrightarrow \Lambda_{t+1}. \quad (10.7)$$

The new bead r^N may alter the interpretation of the sentence still being read. It may cause an earlier passage to be reconsidered, change the likely continuation or open an unexpected neighbouring trajectory. Reading is thus a recurrent process of registration, decompression, comparison, construction and reinjection.

This gives a more exact meaning to the claim that the reader becomes the author. The reader does not merely receive the writer's completed internal object. The writer provides a finite carrier and a constrained sequence of beads. The reader builds a new chain through a different lattice. The written trajectory guides that construction but does not exhaust it.

The operation can be abbreviated as

$$\text{register} \longrightarrow \text{decompress} \longrightarrow \text{compare} \longrightarrow \text{construct} \longrightarrow \text{reinject}. \quad (10.8)$$

The cycle also explains why a later reading of the same page can generate a different result. The marks may remain stable while the handler has changed.

10.5 Phonemes, tokens and new words

At the level of spoken language, a phoneme can be treated as a minimum reusable endogenous linguistic record. It is not asserted to be the smallest physical difference in the acoustic interaction. It is the smallest admitted sound distinction from which a word-chain can be built within a particular linguistic basin. A word may therefore be written as

$$(p_1, p_2, \dots, p_k) \xrightarrow{\kappa_{\Lambda, C}} w, \quad (10.9)$$

where the phonemic beads p_i are compressed into the higher-order word bead w . A familiar word can thereafter be handled without consciously rebuilding each sound. When an unfamiliar word is encountered, however, its phonemic chain can support a first reconstruction before a stable lexical bead exists.

The same architecture recurs at several symbolic scales:

$$\begin{aligned}
 \text{phoneme chain} &\longrightarrow \text{word bead,} \\
 \text{word chain} &\longrightarrow \text{phrase bead,} \\
 \text{sentence chain} &\longrightarrow \text{proposition bead,} \\
 \text{registration chain} &\longrightarrow \text{object bead,} \\
 \text{argument chain} &\longrightarrow \text{theory bead.}
 \end{aligned} \tag{10.10}$$

Written language supplies related but non-identical records through graphemes and character sequences. A contemporary language model commonly operates with tokens or subtokens rather than phonemes. These are implementation-specific beads, not interchangeable linguistic atoms. A token may span several phonemes, while an unfamiliar written word may be divided across several tokens.

This does not prevent the model from handling a new compound or coined word. Before a single vocabulary token exists, the new word can be carried as a chain of already admitted beads. Context can temporarily stabilise the chain as one higher-order functional object. A later training or vocabulary operation may give that compound a more compact label, but functional compression can occur before technical token compression. A system with an explicit phonetic channel can additionally build the new word through its sound-chain.

10.6 Writing and reinjection

Reading need not end with an internal result. A bead generated endogenously can be written, spoken, diagrammed or otherwise impressed into a new external carrier:

$$r^N \xrightarrow{W_{\alpha, \Delta, \delta}} E_{\text{out}}. \tag{10.11}$$

The writing operation W is not the simple reversal of registration. An internal chain must be selected, ordered and compressed into marks that another registration system can distinguish. Some parts of the internal trajectory are given explicit form; others remain unexpressed. The new carrier then begins its own physical and symbolic history.

Reading and writing together form a recurrent loop:

external chain $\longrightarrow$ internal registration $\longrightarrow$ decompression $\longrightarrow$ new endogenous chain $\longrightarrow$ external inscription

(10.12)

The output need not reproduce the input. A reader may compress several pages into one objection, combine the text with an earlier memory, or write a result that was not available before the reading began. The new inscription is a physical record of an endogenous construction, ready to become an exogenously tethered bead in a later reader.

10.7 Information loss and symbolic drag

Symbolic compression is related to information theory, but it cannot be reduced to the number of bits required to encode a sequence. The Theory of Finite Representation must also ask which trajectory was compressed, what licence allows the label to stand for it, what provenance remains attached and which handler will attempt the decompression.

Compression creates a usable bead at a cost. A population becomes a parameter; many detector responses become a value; a page of operations becomes an equation; a changing process becomes a noun. Each compressed bead makes some future operations easier while making parts of the earlier chain difficult or impossible to recover.

Nounification is a particularly strong example. A long and changing trajectory is given a bounded label that can be moved through sentences as though it were a stable object. This is necessary for communication. The difficulty appears when the label is projected backwards and treated as the ready-made cause of the registrations from which it was formed.

Equality can likewise compress two trajectories into one permitted symbolic position. Statistical aggregation compresses many histories into a distribution or parameter. Consensus compresses competing descriptions into a shared vocabulary. In each case the appropriate question is not merely how much information remains, but which differences the present licence has made unavailable.

Symbolic drag is the residual effect of the compressed trajectory upon the new bead. A term carries prior commitments into later sentences; a number carries instrumental conditions into a new calculation; an object carries the history of its classification into a new model. Decompression can restore some of this structure by unfolding provenance, but it cannot recreate a physical event whose registration has been lost.

The later quantitative programme is therefore not simply to calculate a compression ratio. It is to follow compression and decompression along an FST, mark where exogenous provenance entered, identify which distinctions were removed, and examine how different handlers reconstruct the resulting beads. Embedded trajectories may then reveal structure hidden by the compressed endpoints.

Historical note. Robert Greene's 1727 pairing of expansive and contractive forces provides a historical resonance only. The present operations arise from an informational and symbolic framework of finite chains, labels, handlers and reconstruction; no claim is made that Greene possessed or anticipated it.

Compression–decompression principle

Compression licenses one admitted bead to represent a longer chain. Decompression uses that bead to perturb a current handler into reconstructing a further chain. The reconstruction is conditioned by its carrier, provenance, licence and lattice; it need not reproduce the trajectory compressed.

How Objects Become Extant

11.1 From registrations to nouns

An object becomes scientifically extant when a sufficiently stable symbolic compression can be reconstructed from accepted registrations and transformations. The object need not be directly visible to an unaided observer. It must be available within a communal trajectory: named, compared, placed within a class and connected to possible future registrations.

Consider the general sequence

$$\{r_1, r_2, \dots, r_n\} \xrightarrow{F_H} \text{feature} \xrightarrow{L_C} \text{object}. \quad (11.1)$$

The registrations r_i may be spatial samples, repeated responses, instrument states or values collected across time. The FST selects relations among them. A licensing operation then admits the stabilised relation as an object of scientific discourse.

Once formed, the object can organise new measurement. An instrument may be retuned to seek it, a catalogue designed to count it, or a theory extended to predict its behaviour. The symbolic compression feeds back into the physical arrangement. The object is no longer merely an endpoint; it becomes an active constraint upon future trajectories.

11.2 Construction and arbitrariness

To say that an object is constructed is often taken to mean that it is invented without physical constraint. That opposition is too crude. Construction can be tightly tethered to exogenous registration. A bridge may permit only a narrow range of admissible objects, and repeated interactions may resist many proposed classifications.

The word “constructed” should therefore name the positive work through which an object becomes available: selection, distinction, comparison, compression, naming and stabilisation. The physical world participates through the registrations it permits the apparatus to form. The symbolic community participates through the operations it recognises. Neither participation can be erased without distorting the trajectory.

This provides a route between naive copying and unrestricted invention. The scientific object is a stabilised relational achievement. Its reality is examined through the persistence and generativity of its bridges, not through the denial that representation occurred.

11.3 The flying pig and the distant object

The imagined flying pig offers a deliberately simple contrast. A reader can construct a piglet small enough to rest in one hand, with the weight of a tennis ball and webbed wings. Its symbolic object becomes increasingly stable as constraints are added. Different readers will not produce identical mental states, yet their constructions may correspond sufficiently for conversation.

A distant scientific object is also made speakable through a trajectory of constraints. Instrumental values are arranged, segmented, corrected and classified. A stable noun emerges. It would be foolish to conclude that the distant object is therefore imaginary in the same way as the pig. The crucial difference is the direction and provenance of constraint.

The flying pig's trajectory begins predominantly inside a symbolic basin. Its features are licensed by narrative coherence and shared language. The scientific object's trajectory is claimed to begin with exogenous interactions that constrain the registrations. The two processes share symbolic construction but have different bridge structures.

This comparison is valuable because it prevents an easy but false division in which imagination contains construction while observation contains none. Observation must also be reconstructed. The scientific question is whether the reconstruction remains tethered across repeatable bridges.

11.4 Scale and the stability of nouns

Middle-scale objects hide the trajectory of object formation because human perception, action and language have repeatedly stabilised them. A chair remains a chair as one moves around it. Another person can point to it. It can be touched, moved and encountered through several sensory pathways. The noun appears to attach itself directly to the world.

At very small or very large scales, the chain becomes more visible. Ordinary sensory access no longer supplies repeated overlapping registrations. Complex apparatus, processing and mathematical modelling intervene. Familiar nouns such as particle, wave, mass, distance, structure or expansion are extended into regimes where their earlier trajectories may not remain intact.

The problem is not solved by banning nouns. It is addressed by marking their range and preserving the route by which they were licensed. A noun that works at one scale may function as a metaphor, operational placeholder or compressed model at another. Its continuity of spelling does not guarantee continuity of meaning.

11.5 What becomes scientifically extant

Something becomes scientifically extant when it can participate in a stable network of finite symbolic operations. It can be named, counted, compared, predicted or used to organise new registration. This availability depends upon instruments, distinctions, storage, communities and licences. A new apparatus can therefore create new kinds of extant scientific things by making previously unavailable distinctions reproducible.

This does not mean the apparatus creates every physical source of the interaction. It

creates the conditions under which a particular distinction can enter the scientific symbolic world. Before the distinction, there may have been physical processes. After the distinction, there is also a scientific object capable of travelling through papers, models, catalogues and minds.

The old Greene volume provides a small historical counterpart. The work survived because material copies persisted and communities retained enough of the symbolic scheme to recognise them. Its present availability is both physical and symbolic. In reading it anew, another trajectory is formed.

Becoming extant

A scientific object becomes extant when a stable symbolic compression can be reconstructed from admitted registrations and can productively constrain future symbolic or physical operations.

A Provisional Theory

12.1 The complete opening trajectory

The preceding chapters can now be gathered into one provisional architecture:

$$E \xrightarrow{G_{\alpha, \Delta, \delta}} s_0 \xrightarrow{F_H} s_n \xrightarrow{L_C} O. \quad (12.1)$$

Here E is a physical interactional trajectory. The generonic operation G forms a finite symbolic registration s_0 under a distinction condition α , interactional window Δ and uncertainty δ . An endogenous Functional Symbolic Trajectory F_H transforms that registration while carrying an accessible history H . A licensing operation L_C admits the stabilised state s_n as an object, quantity or claim O under conditions C .

Equation (12.1) is not a completed mathematical theory. It is a map of the relations that a completed theory must preserve. Each arrow can contain many nested trajectories. The object O can return as a condition upon a new physical arrangement, and new registrations can reshape the licence under which it was formed. The simple line is a section through a recursive process.

The reverse, predictive trajectory is

$$O \longrightarrow M \xrightarrow{P_C} \hat{\mathcal{R}} \mid_C \mathcal{R}_{\text{new}}, \quad (12.2)$$

where O enters a model M , produces an expected finite representational state and is compared with a new registration. Scientific stability grows as these loops recur without erasing the differences between model, expectation and registration.

12.2 Seven working principles

The opening theory can be stated through seven working principles. They are not intended as commandments imposed upon every discipline. They identify the commitments that define the present trajectory.

1. The Principle of Finite Registration

Empirical access begins when a physical interaction produces a finitely distinguishable and sufficiently stable change. No scientific measurement is available without an interactional

extent.

2. The Axiom of Finite Representation

Every instantiated scientific symbol occupies a finite carrier and has bounded distinguishability, resolution and recoverable provenance, regardless of whether its content refers to an ideal or infinite mathematical object.

3. The Trajectory Principle

No scientific symbol is independent of the transformations through which it was produced. Compression can conceal a trajectory but cannot make the endpoint historically uncaused.

4. The Bridge Principle

An endogenous symbolic construction acquires empirical application only through declared correspondences with possible finite registrations. Internal coherence alone does not create physical identity.

5. The Non-Identity Principle

A representation is not identical with the physical trajectory to which it refers. Strong, repeatable and useful correspondence does not require that difference to be abolished.

6. The Compression Formation Principle

Scientific objects are stabilised by compressing multiple registrations and symbolic trajectories into a representative bead under a licence that declares certain differences irrelevant to the present operation.

7. The Recoverable Decompression Principle

Critical analysis proceeds by decompressing symbolic beads towards their registrations, transformations, uncertainties, provenance and bridge conditions. The reconstruction will always be partial and handler-dependent, but its limits should be declared rather than hidden.

12.3 The status of truth

The Theory of Finite Representation does not need to abandon truth. It changes the form in which a truth claim is carried. A scientific implication can be written

$$P \longrightarrow Q \mid (C, \alpha, H, \delta), \quad (12.3)$$

meaning that the trajectory from P to Q is admitted under consensus or licensing conditions C , a distinction condition α , accessible provenance H and uncertainty δ . These

conditions do not make the claim merely subjective. They state the finite architecture through which the claim can be reconstructed.

An absolute, conditionless sentence may still be formed inside a formal basin. When it is applied to physics, the omitted conditions return through the instrument and bridge. The theory therefore treats empirical truth as a stable and revisable correspondence among trajectories rather than a view from nowhere.

This formulation also avoids making consensus sovereign. The condition C records the licence under which the claim presently travels. New registrations can disturb that licence. A minority trajectory can produce a stronger bridge. Consensus stabilises a symbolic object; it does not place the object beyond interaction.

12.4 Admissibility rather than prohibition

The theory is most useful when it asks whether a move is admissible under declared conditions, not when it issues universal prohibitions. A continuum may be admissible in a mathematical proof and productive in a physical model. The question is which finite relations support its use and which physical claims extend beyond those relations.

Similarly, an exact equality can be admissible in algebra while a measured comparison requires an interactional bar. A statistical population can be admissible for prediction while concealing the individual trajectories from which it was formed. A noun can be admissible for communication while becoming misleading when projected across scale.

Admissibility keeps alternative pathways open. It permits us to explore what happens when finite extent, provenance or different bridge rules are placed at the beginning. The resulting theory must still earn its bridges through registration. It is not protected by novelty.

12.5 What the theory does not yet claim

This first volume does not provide a universal metric for symbolic distance, a complete calculus of compression and decompression or a final physical value for the Alphonic limit. It does not prove that all continua are physically false, that all scientific objects are reducible to language, or that consensus alone creates reality. It does not claim that Greene anticipated Finite Symbolic Mechanics.

These absences are not evasions. They mark the present extent of the work. The theory has established an architecture: three basins, a registration boundary, five conditions of representation, finite trajectories, bridge licensing and a compression–decompression dynamics. Later work can test, alter and extend that architecture without pretending that its next results were contained in the opening axioms.

Provisional statement of the theory

Scientific knowledge develops as a recurrent coupling between finite physical registrations and finite symbolic trajectories. Objects and claims become extant through licensed compressions; inquiry proceeds by producing new distinctions and by decompressing the histories that compression has concealed.

Directions of Travel

13.1 Language

Language is not a transparent carrier placed around already completed thought. Words divide, gather and redirect symbolic space. Nouns contract processes into objects; verbs unfold relations across time; metaphor transports constraints between basins. A theory of finite representation can examine these operations as trajectories rather than treating meaning as a static correspondence between word and thing.

This direction includes the historical question of how particular symbolic licenses became dominant. The language of seventeenth-century classification, mathematics and natural philosophy did not merely report a new scientific world. It helped create the routes through which that world became speakable. Recorde's equality, Euclidean objects, institutional publication and the authority of named disciplines belong to this history of licensing.

13.2 Mathematics and proof

A proof can be unfolded as a finite symbolic trajectory. Its apparently static page then becomes an ordered sequence of admissible transitions. Different proofs of one result may approach different stabilising regions, preserve different structures or depend upon different bridge theorems.

Embedding these trajectories offers a practical programme. Parameters can be stepped, states enumerated and delayed coordinates constructed. The resulting attractor is not merely an illustration of the proof. It is a representation of the proof's functional dynamics under the chosen unfolding. Questions of reachability, stability and non-reachability can then supplement conventional symbolic judgement.

The finite approach does not need to begin by defeating classical mathematics. It can build alternative endogenous games from different commitments and compare their trajectories and physical bridges.

13.3 Instruments and conversion

Instrumentation makes the registration boundary materially accessible. A sensor responds, a signal stabilises, an ADC assigns codes, software transforms them, and a display offers

a new symbolic surface. Each stage can be examined for its distinction scale, uncertainty, provenance and compressive operation.

The ADC is especially important because it performs an explicit crossing from a range of physical responses to a finite symbolic alphabet. Yet it should not be isolated from the sensor before it or the software after it. The complete generonic chain includes the conditions under which the analogue response was formed and the later operations through which the code becomes a measurement.

Experiments with light, pinholes and CCD registration can investigate how patterns stabilise across finite windows. Diffraction rings, near-field responses and changes of detector configuration offer ways to embed sequences rather than treating each image as a timeless object. These experiments need not be made the foundation of the general theory; they can become tests of its physical basin and bridge rules.

13.4 Images and astronomical objects

A scientific image appears immediate because human vision compresses its structure rapidly into objects. But the displayed colours, intensities and boundaries often arise from processing choices that are not themselves visible in the image. To ask what an image is “of” is to ask about its provenance and bridge.

Astronomy offers a particularly long trajectory from remote interaction to scientific noun. Incoming interactions change a detector; values are calibrated and arranged; sources are segmented; objects are catalogued; populations are modelled; cosmological statements are formed. The Vera C. Rubin Observatory could provide a powerful future case study because of the scale and visibility of this pipeline. It should remain one case among many rather than becoming the boundary of the theory.

13.5 Statistics and trajectory loss

Statistics gains power by compressing individual histories into collective structure. A distribution can reveal regularity while making the order and path of its members unavailable. The Theory of Finite Representation can ask which trajectory information survives each statistic and whether different dynamics have been collapsed into similar endpoints.

This provides a route into the relation between entropy and lost history. Instead of treating entropy only as an abstract quantity attached to a set of states, one can examine the finite operations through which their generating paths became inaccessible. Embedding may recover some structure, while other losses remain irreversible.

13.6 Artificial intelligence

A language model operates upon finite tokens, vectors and transformations, yet its symbolic trajectories can reconstruct patterns extending far beyond any single token. Attention can be examined as a form of relational delay embedding, and a Takens-based transformer can explore whether trajectory reconstruction offers an alternative architecture.

The theory also changes the alignment question. If language is treated as a transparent vessel, alignment becomes the correction of outputs against externally chosen rules. If language is a nonlinear dynamical system of finite trajectories, alignment requires an account of the basins, attractors and licences through which meanings stabilise. The model and its human reader form a coupled trajectory rather than a one-way transfer of finished propositions.

13.7 The programme of Part II

Part II should follow these crossings comparatively. Its organising subject need not be one telescope, one equation or one branch of physics. It can study the dynamics of finite representation across domains while maintaining the three-basin separation established here.

A provisional title is

A Theory of Finite Representation

Part II: Compression, Decompression and the Formation of Objects.

The next volume would seek operational measures of symbolic compression and decompression, methods for embedding FSTs, accounts of trajectory loss, comparative instrument studies and extended examples of noun formation. The present volume has supplied the doorway and the commitments under which that travel can begin.

Epilogue: Another Trajectory Begins

The old book remains on the desk. Its title offers expansive and contractive forces; its page calls certain rational sciences extant; its material surface carries a history that cannot be fully read. The volume has moved through physical hands and symbolic communities until it has entered this argument.

Nothing in the present work requires Robert Greene to have imagined a Theory of Finite Representation. The connection was made by a new reader under new conditions. That is not a weakness in the chain. It is what reading does. A past symbolic trajectory is reconstructed, altered by the present basin and carried towards another possible future.

The same is true of mathematics and science. We inherit stable paths. They have earned their authority through centuries of successful reconstruction and application. Yet inherited success does not establish that no alternative path can be taken. A licence creates a domain of work; it does not exhaust the landscape.

Finite representation begins with this freedom and this responsibility. We can choose another starting commitment, but the resulting trajectory must remain inspectable. We must say where distinctions entered, which histories were preserved, what uncertainty was carried, what compression formed the object and what bridge licensed the return to physics.

A finite representation does not terminate the world in a symbol. It creates a bounded and historically situated place from which another trajectory can begin.

Working glossary

Admissibility

The conditions under which a representation or transition is accepted for a particular operation within a symbolic or scientific community.

Alphonic distinction

The operative boundary at which a relevant difference is registered as non-single and becomes available to a symbolic trajectory.

Bridge

A declared mapping or correspondence between a physical interactional trajectory and a functional symbolic trajectory.

Bead

A finite symbolic distinction capable of occupying a position in a chain, carrying provenance and conditioning a later operation.

Compression

An operation through which one admitted bead is licensed to represent a longer chain of beads under stated conditions.

Decompression

The handler-dependent reconstruction of a further bead-chain from a compressed symbolic bead. The reconstructed chain need not be identical to the trajectory originally compressed.

Extant

Available to be encountered and reconstructed within a present physical and symbolic system.

Finite extent

The non-zero interactional span across which a physical distinction is produced and stabilised.

Finite representation

An instantiated, bounded and historically situated distinction capable of conditioning a later symbolic operation.

Functional Symbolic Trajectory

A finite pathway of symbolic states and transformations through which constraints, uncertainty, meaning and admissibility are carried.

Generonic boundary

The functional region in which a physical interaction becomes available as a finite symbolic registration.

Handler

The current endogenous lattice through which an incoming bead or chain is compared, decompressed and incorporated into a continuing trajectory.

Interactional bar

A notation for conditioned correspondence that preserves the distinction between related trajectories rather than declaring their identity.

Licence

A set of commitments and permissions that makes some symbolic trajectories admissible and excludes others.

Provenance

The accessible portion of the physical, symbolic and institutional history through which a representation became present.

Registration

A finite relational event in which physical interaction produces a sufficiently stable distinction to condition a later symbolic state.

Symbolic drag

The residual dependence of a compressed symbol upon the trajectory and commitments through which it was produced.

References and source directions

This draft is an internally developed conceptual monograph rather than a literature review. A later scholarly edition should supply full primary citations and critical engagement. The following entries mark the principal historical and theoretical directions already present in the argument.

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